

LAB EXPERIMENT – 5

Missionaries and Cannibals Problem using Python — CO3

Aim

To write a Python program to solve the **Missionaries and Cannibals problem** using a **Breadth-First Search (BFS)** technique.

Procedure

1. Consider **3 missionaries and 3 cannibals** on the left side of a river.
2. A boat can carry a maximum of **2 people**.
3. At least one person must be in the boat during a crossing.
4. Cannibals must never outnumber missionaries on either side when missionaries are present.
5. Represent a state as:
(Missionaries_Left, Cannibals_Left, Boat_Position).
6. Generate all possible valid boat movements.
7. Use BFS to search for a safe path.
8. Continue until all missionaries and cannibals reach the right side.
9. Display the solution steps.

Source Code

main.py

Visualize

Run ▶

```
1 from collections import deque
2
3 def is_valid(m_left, c_left):
4     m_right = 3 - m_left
5     c_right = 3 - c_left
6
7     # Values must be within limits
8     if not (0 <= m_left <= 3 and 0 <= c_left <= 3):
9         return False
10
11     # Left side condition
12     if m_left > 0 and m_left < c_left:
13         return False
14
15     # Right side condition
16     if m_right > 0 and m_right < c_right:
17         return False
18
19     return True
20
21
22 def solve():
23     start = (3, 3, 0) # 0 = Left, 1 = Right
24     goal = (0, 0, 1)
25
26     # Possible boat combinations
27     moves = [
28         (1, 0),
29         (2, 0),
30         (0, 1),
31         (0, 2),
32         (1, 1)
33     ]
34
35     queue = deque()
36     queue.append((start, [start]))
37
38     visited = {start}
39
40     while queue:
41         state, path = queue.popleft()
42
```

Output

Missionaries and Cannibals Solution:

Step 0: Missionaries Left = 3, Cannibals Left = 3, Boat = Left
Step 1: Missionaries Left = 3, Cannibals Left = 1, Boat = Right
Step 2: Missionaries Left = 3, Cannibals Left = 2, Boat = Left
Step 3: Missionaries Left = 3, Cannibals Left = 0, Boat = Right
Step 4: Missionaries Left = 3, Cannibals Left = 1, Boat = Left
Step 5: Missionaries Left = 1, Cannibals Left = 1, Boat = Right
Step 6: Missionaries Left = 2, Cannibals Left = 2, Boat = Left
Step 7: Missionaries Left = 0, Cannibals Left = 2, Boat = Right
Step 8: Missionaries Left = 0, Cannibals Left = 3, Boat = Left
Step 9: Missionaries Left = 0, Cannibals Left = 1, Boat = Right
Step 10: Missionaries Left = 1, Cannibals Left = 1, Boat = Left
Step 11: Missionaries Left = 0, Cannibals Left = 0, Boat = Right

```
main.py Visualize Run ▶ Output
43
46     m, c, boat = state
47
48     for dm, dc in moves:
49
50         if boat == 0: # Boat on left
51             new_m = m - dm
52             new_c = c - dc
53             new_boat = 1
54         else: # Boat on right
55             new_m = m + dm
56             new_c = c + dc
57             new_boat = 0
58
59         new_state = (new_m, new_c, new_boat)
60
61         if is_valid(new_m, new_c) and new_state not in visited:
62             visited.add(new_state)
63             queue.append(
64                 (new_state, path + [new_state])
65             )
66
67     return None
68
69
70 solution = solve()
71
72 if solution:
73     print("Missionaries and Cannibals Solution:\n")
74
75     for i, state in enumerate(solution):
76         m, c, boat = state
77
78         side = "Left" if boat == 0 else "Right"
79
80         print(
81             f"Step {i}: "
82             f"Missionaries Left = {m}, "
83             f"Cannibals Left = {c}, "
84             f"Boat = {side}"
85         )
86     else:
87         print("No solution exists.")
```

Missionaries and Cannibals Solution:

Step 0: Missionaries Left = 3, Cannibals Left = 3, Boat = Left
Step 1: Missionaries Left = 3, Cannibals Left = 1, Boat = Right
Step 2: Missionaries Left = 3, Cannibals Left = 2, Boat = Left
Step 3: Missionaries Left = 3, Cannibals Left = 0, Boat = Right
Step 4: Missionaries Left = 3, Cannibals Left = 1, Boat = Left
Step 5: Missionaries Left = 1, Cannibals Left = 1, Boat = Right
Step 6: Missionaries Left = 2, Cannibals Left = 2, Boat = Left
Step 7: Missionaries Left = 0, Cannibals Left = 2, Boat = Right
Step 8: Missionaries Left = 0, Cannibals Left = 3, Boat = Left
Step 9: Missionaries Left = 0, Cannibals Left = 1, Boat = Right
Step 10: Missionaries Left = 1, Cannibals Left = 1, Boat = Left
Step 11: Missionaries Left = 0, Cannibals Left = 0, Boat = Right

Result

Thus, the **Missionaries and Cannibals** problem was successfully solved using Python and **BFS**.